

Mathematics 3A03 Real Analysis I

Extra Problems

Topic: **Mainly Metric Spaces**

Miscellaneous

1. Prove that if $f(0) = 0$ and $f(x) = x^2 \sin \frac{1}{x}$ otherwise, then f is differentiable at 0 but f' is not continuous in any interval containing 0.

2. (**Weierstraß function**) Let

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

where $0 < a < 1$, $b \in \mathbb{N}$ is odd, and $ab > 1$. Prove that W is

- (a) continuous everywhere;
- (b) differentiable nowhere.

Hint: $\cos(A+B) - \cos A = -2 \sin\left(\frac{B}{2}\right) \sin\left(A + \frac{B}{2}\right)$. Also, recall that to prove that a function $f(h)$ does not converge as $h \rightarrow 0$, it is sufficient to find a sequence (h_n) such that $h_n \xrightarrow{n \rightarrow \infty} 0$ but the sequence $(f(h_n))$ does not converge as $n \rightarrow \infty$.

Metric Spaces

3. Suppose $\mathcal{M}$ is a set, a is a real number, and for all $x, y \in \mathcal{M}$,

$$d_a(x, y) = \begin{cases} 0 & x = y, \\ a & x \neq y. \end{cases}$$

For what values of $a \in \mathbb{R}$ is $(\mathcal{M}, d_a)$ a metric space? Does the answer depend on $\mathcal{M}$?

4. Let $\mathcal{M}$ be a (non-empty) finite set, so $\mathcal{M} = \{x_1, \dots, x_n\}$ for some $n \in \mathbb{N}$. What are all the possible metrics on $\mathcal{M}$? (Start with $n = 1$, then 2, then 3 and then try to describe the situation for general n . *Warning:* I'm not aware of a simple algebraic condition in general. If you find one, please let me know.)

5. Let $\mathbb{R}^\infty$ denote the space of sequences of real numbers. For $1 \leq p < \infty$, consider the subspace

$$\ell^p = \left\{ x = (x_1, x_2, \dots) \in \mathbb{R}^\infty : \sum_{i=1}^{\infty} |x_i|^p < \infty \right\}$$

with norm

$$\|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p \right)^{1/p}.$$

Prove that the closed unit ball in ℓ^p ,

$$\mathcal{B}_p = \left\{ x \in \ell^p : \|x\|_p \leq 1 \right\},$$

is not compact.

6. Consider ℓ^∞ , the space of bounded sequences of real numbers with the sup norm, as defined in class on slide 70 of the metric spaces lectures. Let $E = (0, 1)^\infty$ be the subset of ℓ^∞ (as in the example in class) of sequences that are strictly in $(0, 1)$. Give examples of points in E° and ∂E . Prove that E is neither open nor closed, and find E° and $\overline{E}$. Is $\overline{E}$ compact?

7. Suppose V is a vector space and $f : V \rightarrow \mathbb{R}$. Prove or disprove each of the following statements.

- (a) If f satisfies the parallelogram law,

$$f(x+y)^2 + f(x-y)^2 = 2f(x)^2 + 2f(y)^2, \quad \forall x, y \in V,$$

then $f(0) = 0$.

- (b) If f satisfies the parallelogram law, then the function g , defined by $g(x) = (f(x))^2$ for all $x \in V$, also satisfies the parallelogram law.
 (c) If f satisfies the parallelogram law, then f is a norm on V .
 (d) If f satisfies the parallelogram law, then $d(x, y) = f(x - y)$ is a metric on V .

8. (a) Prove that the zero function, $f(x) = 0$ for all x , is the unique solution to the integral equation

$$f(x) = \int_0^x f(t) dt \quad \forall x \in [0, \tfrac{1}{2}].$$

Hint: Consider an appropriate metric space and use the Contraction Mapping Principle.

- (b) Reconsider the result in part (a) but with the interval on which f is defined being $[0, b]$. For which $b \in (0, 1] \setminus \{\frac{1}{2}\}$ is the result in part (a) true?

9. Prove or disprove: A contraction map on a metric space is continuous.
10. Prove that any two norms on $\mathbb{R}^n$ are **equivalent** in the sense that $\exists C_1, C_2 > 0$ such that

$$C_1 \|x\|_a \leq \|x\|_b \leq C_2 \|x\|_a, \quad \forall x \in \mathbb{R}^n.$$

Hint: Consider the unit sphere (the boundary of the unit ball).

Does it follow that all norms on $\mathbb{R}^n$ are equivalent? What can you infer about the topology (class of open sets) induced by norms on $\mathbb{R}^n$?

Denote the space of *eventually zero* sequences by $\mathbb{EZ}$ (think “eee-zee”). Are all norms on $\mathbb{EZ}$ equivalent? Is this an easy problem?

11. Prove that $\mathbb{R}^n$ is complete with respect to (a) the Taxicab norm ($\|\cdot\|_1$) and (b) the Maximum ($\|\cdot\|_\infty$) norm.

In fact, for any $p \geq 1$, $\mathbb{R}^n$ with norm $\|\cdot\|_p$ is complete with respect to the metric $d(x, y) = \|x - y\|_p$ induced by the norm.

Is $\mathbb{R}^n$ complete with respect to any norm on $\mathbb{R}^n$?

Construction of $\mathbb{R}$

Note: This topic was not covered in class in 2026 so it will not come up on the exam in 2026. However, my slides for a lecture on this topic are on the course website after all the lectures that actually occurred. If you read those slides (which I encourage you to do!) then the following problem will help solidify your understanding.

12. Recall from class that we defined a **real number** to be a subset $\alpha \subseteq \mathbb{Q}$ with the following four properties:

1. **downward closed:** $\forall x \in \alpha$, if $y \in \mathbb{Q}$ and $y < x$, then $y \in \alpha$;
2. **non-empty:** $\alpha \neq \emptyset$;
3. **non-empty complement:** $\mathbb{Q} \setminus \alpha \neq \emptyset$, i.e., $\alpha \neq \mathbb{Q}$;
4. **no greatest element:** $\forall x \in \alpha$, $\exists y \in \alpha$ such that $y > x$.

Assume α and β are real numbers, and define their **sum** $\alpha + \beta$ to be

$$\alpha + \beta \stackrel{\text{def}}{=} \{a + b : a \in \alpha, b \in \beta\}.$$

Use the formal definition above to show that $\alpha + \beta$ is a real number.

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